A simulation model for thermal performance prediction of a coal-fired power plant

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Abstract

This study deals with the simulation modeling of a 250 MW capacity coal-based power plant at different load conditions. The plant under study is subcritical, reheating and regenerative type situated in northern part of India. At present, computer-based programming and advanced software used in simulation, modeling and optimization used in process industries playing an important role. The present work related with the semi-empirical modeling for thermal performance analysis of a coal-fired power plant. The model basically includes the mass, energy balance equations. A typical thermodynamic property relation with empirical correlations using steam bled pressure extracted from turbine at various points in plant has been used. The necessary data collected from plant and model are applied in a general programming form for plant analysis. The process-based simulation model with the integration of steam generator, turbine, pump, condenser and feedwater heaters in the plant has been successfully run using the MATLAB calculation tool. The validation of the model has been done by comparing overall plant efficiency, coal consumption rate and mass flow rate of steam at various load conditions for 250 MW capacity plant under operating conditions. The results achieved through simulation model are found to be satisfactory. It is expected that this simulation model will help out to those engineers and researchers who are dealing with the improvement of design, analysis and optimization of coal-fired power plant.

Keywords: thermodynamic modeling; thermal power plant; plant efficiency; steam turbine; computational algorithm

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1 INTRODUCTION

Today's electric energy is going to prove a better option for the development of any country and it seems to be impossible to leave without it. The people are totally dependent on electricity for their daily needs in developing as well as in developed country. Coal is the main source of electricity generation in fossil fuel based power plants. The use of coal in coal-fired power plant (CFPP) causes raise in global temperature due to greenhouse gases emission responsible for global warming across the globe [1, 2]. It also has an adverse impact on the health of human being. Also, a huge amount of water is required for CFPP and the lack of freshwater resources to meet this demand is another big problem in many parts of the world [3]. The entire world is facing energy-related issues today and searching the various options to resolve it as soon as possible. The uncontrollable population of any nation making it difficult to fulfill their daily energy demand. Due to population growth and improvement in advance technology, energy is going to be consumed at a faster rate [4]. Moreover, a major amount of energy is going to be consumed to run various project in different fields necessary for a nation development. Energy systems play an important role in any nation economy. To fulfill this need, in a number of power plants are working in a continue manner and in a number of researchers are trying to optimize these various power plants by using new advanced technology to

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fulfill their present countrymen needs. Coal-fired power plants are playing a crucial role in this area. Its working is to convert coal stored energy to usable electricity. Coal-based power plants contribute total 80% of power requirements to world and remaining demand is contributed by other sources of power [5]. Today's world energy crisis and environmental changes are going to provide various opportunities to search out various energy efficient advanced technologies [6]. Researchers are applying their efforts to improve the efficiency of the power plant. A new approach for feedwater heater train simulation to increase overall plant efficiency in a CFPP was designed and fruitful results were obtained [5–7]. In a number of researcher have focused their research on physical and chemical processes taking place in boiler combustion chamber to enhance boiler as well as plant efficiency in thermal power plants [8–12]. A simplified simulation model was proposed to predict the performance of a circulating fluidized bed boiler under varying loads conditions [13]. Rankine cycle power plant efficiency can be increased using a combined cycle [13, 14]. Also a supercritical boiler model was compared using fuzzy-neural network and recursive least squares method to observeboiler efficiency [15]. A nonlinear mathematical model to characterize the transient dynamics at various points for a steam turbine was proposed [16]. Mathematical programming algorithms were imported in software to optimize overall CFPP efficiency [15, 16]. Thermodynamic optimization of gas turbine based plants is also suggested [17, 18]. The incorporation of solar energy with power plants is also playing an important role nowadays [19]. Ahmadi *et al.* [20] proposed a novel system to recover the wasted heat and water along with exergy and economic analyses in thermal power plant. India's electricity sector itself also uses in majority the total coal produced in country. In a number of CFPP's are using subcritical steam parameters in majority and only a couple of plants are utilizing supercritical steam parameters in India. Thermal power plants performance can be evaluated all the way through energetic performance criterion, which is electrical power and thermal effectiveness. Energy is an essential tool for the development of present as well as future economy of a nation. The plant efficiency can be improved using modification in plant configuration or with optimization in plant operating conditions. So, in present study simulation model has been proposed to see the effect of various parameters on plant thermal performance.

2 POWER PLANT CYCLE DESCRIPTION

This paper presents simulation modeling of a subcritical CFPP of 250 MW capacity situated in North India. The plant with all components detail is shown in Figure 1. In steam generator (SG), steam is generated through the chemically treated water circulation all around water tube walls of SG furnace. Superheater

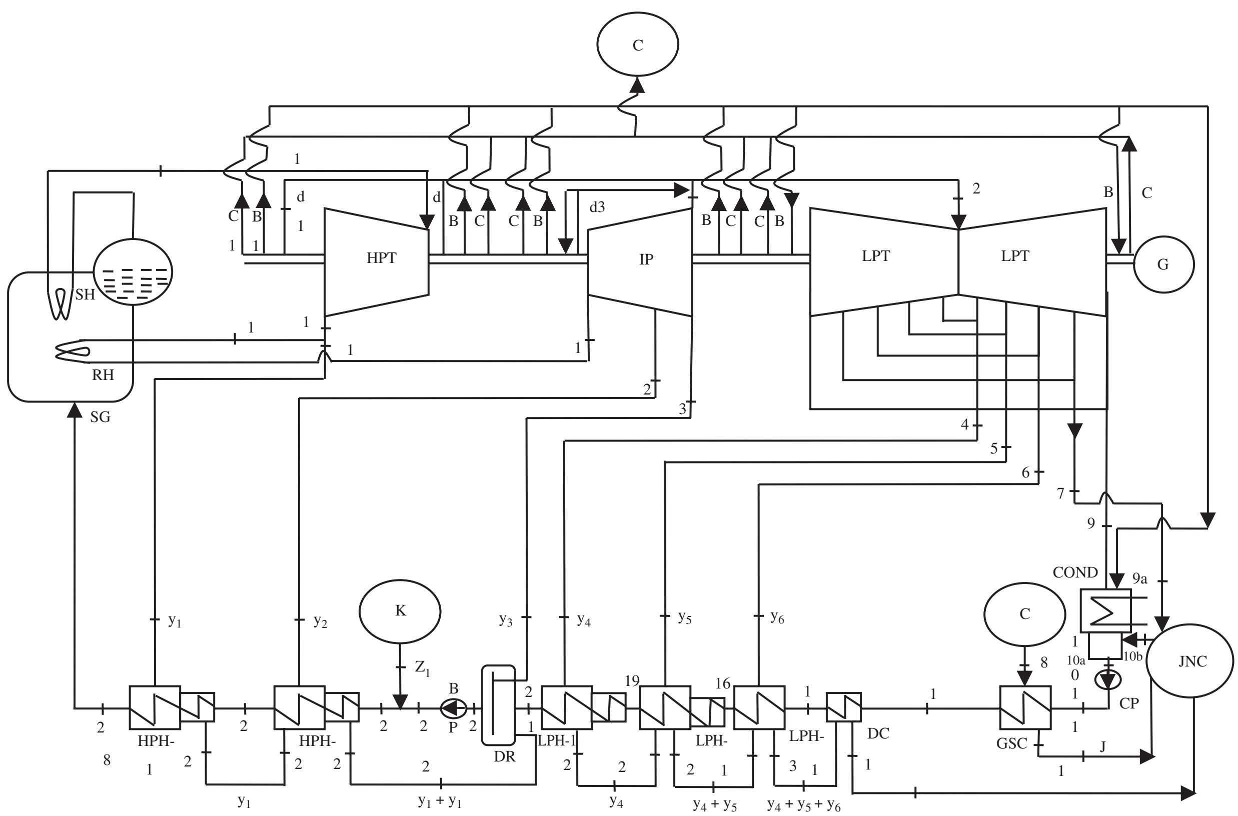

Figure 1. *Steam turbine cycle based coal-fired power plant 250 MW.*

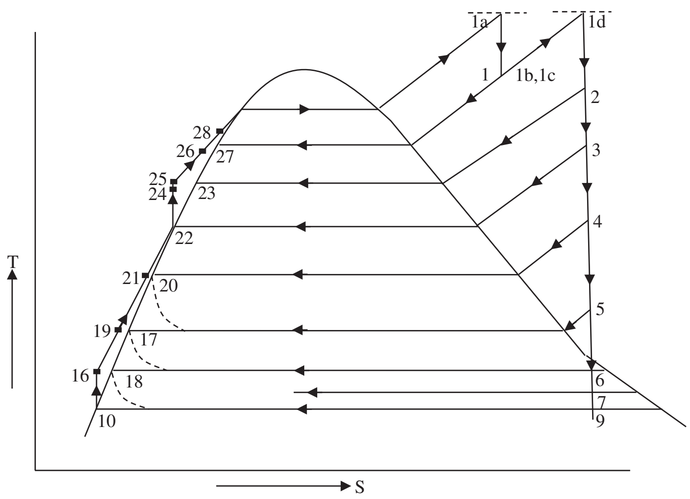

Figure 2. *Temperature-entropy (T-s) diagram of plant.*

(SH) increases the temperature of steam from SG up to 537°C and then it enters into high pressure turbine (HPT). After doing useful work in HPT steam enters in reheater (RH) to increase steam temperature from 344°C to 537°C and then enters in intermediate pressure turbine (IPT). Also, some part of bled steam is allowed to condense in various feedwater heaters as well. After that, the balance steam enters in the low pressure turbine at a temperature of 301°C and finally this bled steam is collected at four stages (see Figure 1), while last bled steam is introduced in the condenser unit. At the exit of condensate extraction pump (CP) condensate enters in series of feedwater heaters. In low pressure heaters, feedwater is heated up by the steam extracted from the turbines. The water leaving low pressure heaters enters the deaerator. Boiler feed pump ejects this water into high pressure heaters. In high pressure heaters, feedwater is heated up again. Finally, it enters in Steam generator at a higher temperature passing through economizer. Temperature-entropy (T-s) diagram of plant is shown in Figure 2.

3 THERMODYNAMIC MODELING

In this paper, the main focus is on the simulation modeling along with the integration of all necessary components responsible for power plant working (see Figure 3). So, mathematical modeling of each plant component using empirical equations related to mass and energy balance is represented in this work. The extracted bled steam pressure from turbine to feedwater heaters has been also used. In this analysis, the input and output values of the plant components are determined using the measured or calculated thermodynamic variables such as enthalpy, pressure, temperature, entropy, mass flow rate and quality.

3.1 Steam generator

The function of the steam generator (Boiler) is to generate the steam of desired quality. Generally two methods are used for

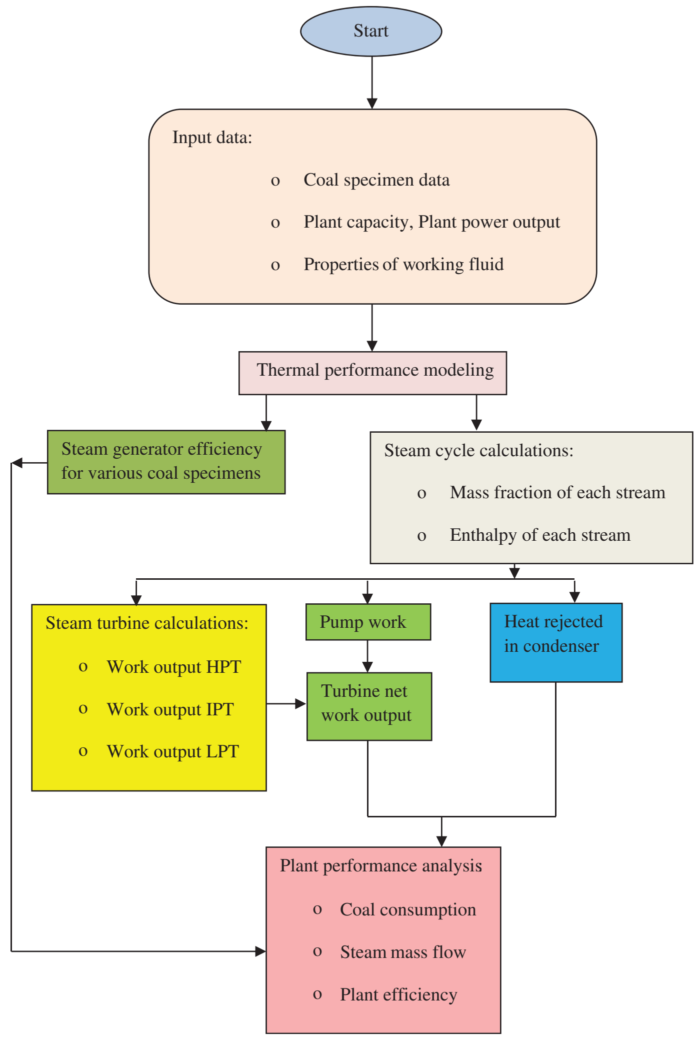

Figure 3. *Computational algorithm of plant simulation model.*

the evaluation of SG efficiency namely direct and indirect method. In this work, indirect method has been adopted for the evaluation of SG efficiency (see Table 1). Four coal samples have been chosen from the plant (see Table 2) to see their impact on SG as well as on plant efficiency.

Radiation and convection loss is taken as 0.21% [21]. Energy and mass balance equations used for other plant components used is shown in Table 3. Working fluid properties modeling can be easily obtained [22, 23].

3.2 Feedwater heaters and condensers

For formulation, feedwaters are considered to be adiabatic. The entry temperature difference (ETD) and terminal temperature difference (TTD), it is also assumed to be non-zero and constant quantity for each feedwater heater irrespective to change in power output. Pressure drop across the feedwater heaters is assumed to be negligible.

Table 1. *Steam generator efficiency in plant using losses method.*

Steam generator loss	Expression	Unit
Loss due to dry flue gas (L_1)	$[\frac{100}{12(CO_2+CO)}(\frac{C}{100}+\frac{S}{267}-Ca)]30.6(T_f-T_a)$ (1)	kJ/kg of fuel
Loss due to wet flue gas (L_2)	$(\frac{M+9H}{100})[1.88(T_f-25)+2442+4.2(25-T_a)]$ (2)	kJ/kg of fuel
Heat loss due to moisture present in fuel (L_3)	$L_2-(Coal_{GCV}-Coal_{NCV})$ (12)	kJ/kg of fuel
Heat loss due to moisture present in air (L_4)	$1.88m_a c_p(T_f-T_a)$ (3)	kJ/kg of fuel
Unburnt gas loss (L_5)	$[\frac{CO}{(CO_2+CO)}(\frac{C}{100}+\frac{S}{267}-Ca)]23717$ (4)	kJ/kg of fuel
Radiation and convection loss (L_6)	0.21	%
Steam generator efficiency (η_b)	$[100-(\sum_{i=1}^{5}L_i\times 100/Coal_{GCV})-L_6]$ (5)	%
Here, actual mass of air supplied per kg of coal(m_a)	$[\frac{3.034N}{(CO_2+CO)}(\frac{C}{100}+\frac{S}{267}-Ca)]$ (6)	

Table 2. *Proximate and ultimate analysis of various coal samples (*).*

	Proximate analysis				Ultimate analysis					
Coal specimen	Moisture content (M)	Volatile matter (VM)	Ash content (A)	Fixed carbon (FC)	Oxygen (O) in %	Sulfur (S) in %	Carbon (C) in %	Hydrogen (H) in %	Nitrogen (N) in %	$Coal_{GCV}$ (kJ/kg)
Coal specimen-1	9.7	25.7	27.0	37.6	7.07	0.60	48.46	3.44	1.03	4530
Coal specimen-2	8.1	20.7	36.0	35.2	4.27	0.51	43.51	3.03	0.98	3960
Coal specimen-3	6.5	24.5	41.3	27.7	6.68	0.55	37.15	2.83	0.86	3520
Coal specimen-4	7.1	20.4	48.9	23.6	5.55	0.24	30.82	1.90	0.60	2670

*Data collected from plant of coal specimen.

Table 3. *Energy and mass balance equations used in plant.*

Mass balance equation	$\sum\dot{y}_{in}=\sum\dot{y}_{out}$ (7)	Power consumed by pumps	$\dot{W}_p=\frac{\dot{y}_{in}(h_{out}-h_{in})}{\eta_p}$ (10)
Energy balance equation	$\dot{Q}-\dot{W}=\sum\dot{y}_i(h_{out,i}-h_{in,i})$ (8)	Net electrical power output	$\dot{W}_{Net}=\sum\dot{W}_T-\sum\dot{W}_P$ (11)
Turbine work output	$\dot{W}_T=\dot{y}\Delta h$ Where, $\Delta h=h_{in}-h_{out}$ (9)		

3.2.1 *High pressure feedwater heater (HPH-1)*

The schematic diagram of HPH-1, which corresponds to three zones including desuperheating, condensing and drain-cooling zone (the extracted steam upon condensation gets subcooled, which needs to be drain out using a drain cooler, DC) has been shown in Figure 4. HPH-1 receives superheated steam bled from the turbine at state 1, the steam is first desuperheated, then condensed and finally subcooled at state 27, whereas the feedwater gets heated from state 26 to 28. Since, the feedwater heater is a closed type heat exchanger with finite surface area; the feedwater temperature usually differs from the condensing temperature of the extraction stream. Since the information of complete geometry of feedwater heaters (heat exchangers) was not available, in such situation, two parameters were defined for each unit namely TTD and ETD, respectively. TTD is defined as the difference between saturated temperature of the bled stream and exit water temperature, while ETD is defined as the difference between saturated temperature of bled steam and inlet water temperature.

Mathematical modeling comprises of energy and mass balance for HPH-1 with respect to the unit feedwater flow rate entering into the heater and 'i' = 1 as shown in Figure 5.

Energy balance:

$$y_i\times h_{s(i)}+1\times h_{f(i+1)}=1\times h_{f(i)}+y_i\times h_{c(i)} \quad (12)$$

$$y_i=(h_{f(i)}-h_{f(i+1)})/(h_{s(i)}-h_{c(i)}) \quad (13)$$

where

$$h_{f(i)}=h_{(i)l}-TTD\times Cpw \quad (14)$$

$$h_{c(i)}=h_{f(i+1)}+ETD\times Cpw \quad (15)$$

Here 'y' is the fractional mass flow with total steam flow from steam generator. Cpw is specific heat capacity of working fluid. TTD and ETD are terminal and ETD of feedwater heaters. The

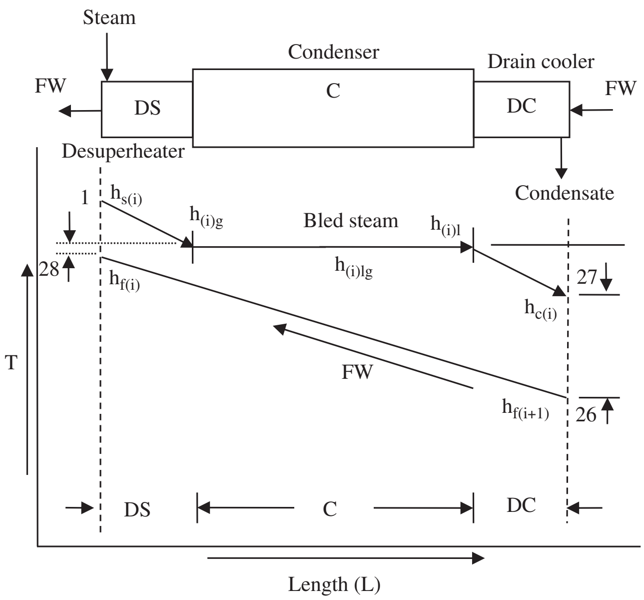

Figure 4. *Temperature–length diagram for HPH-1.*

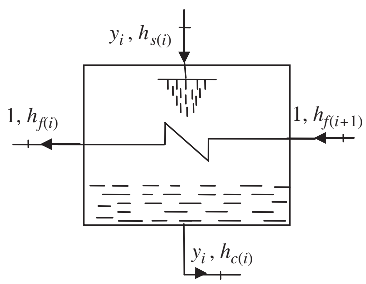

Figure 5. *Energy balance diagram for high pressure heater (HPH-1).*

data of TTD and ETD for each feedwater heaters are listed in Table 4.

The bleeding steam pressure extracted from turbine to feedwater heaters at various points (see Table 5) is given by equation below

$$p_i = m_i MW + n_i \tag{16}$$

The constants values in equation (16) are given (see Table 6). The values of these constants are obtained through the curve fitting of bled steam extraction pressure from turbine to feedwater heaters at various points in cycle.

3.2.2 *High pressure feedwater heater (HPH-2)*

Physically, HPH-2 is similar to HPH-1, except it receives one additional steam bled of condensate leaving from HPH-1 as shown in Figure 6. Heater receives superheated steam bled from the turbine at state 2, the steam is first desuperheated, then condensed and finally subcooled at state 23, whereas the feedwater gets heated from state 25 to 26 (refer Figure 6).

In the same manner for HPH-2, the subscript '*i*' is fixed at 2, Energy balance:

$$y_i \times h_{s(i)} + 1 \times h_{f(i+1)} + y_{i-1} \times h_{c(i-1)} = 1 \times h_{f(i)} + (y_i + y_{i-1}) \times h_{c(i)} \tag{17}$$

$$y_i = (h_{f(i)} - h_{f(i+1)})/(h_{s(i)} - h_{c(i)}) + y_{i-1}((h_{c(i)} - h_{c(i-1)})/(h_{s(i)} - h_{c(i)})) \tag{18}$$

where

$$h_{f(i)} = h_{(i)l} - \text{TTD} \times Cpw \tag{19}$$

$$h_{c(i)} = h_{f(i+1)} + \text{ETD} \times Cpw \tag{20}$$

Table 4. *Terminal temperature and ETD of feedwater heaters.*

Feedwater heaters	ETD	TTD
LPH-1	5.0	6.0
LPH-2	4.0	2.0
LPH-3	6.0	5.0
HPH-1	3.0	6.0
HPH-2	4.0	5.0

Table 5. *Bled steam pressure (bar) from turbine to feedwater heaters at various points in design conditions.*

Sr. no.	MW	P_1	P_2	P_3	P_4	P_5	P_6
1	100	16.86	7.29	3.12	1.099	0.423	0.19
2	125	20.49	8.74	3.74	1.316	0.509	0.231
3	150	24.9	10.5	4.26	1.497	0.583	0.264
4	175	28.52	11.99	4.86	1.703	0.665	0.297
5	200	32.2	13.51	5.47	1.911	0.474	0.328
6	250	39.67	16.57	6.68	2.332	0.911	0.385

Table 6. *Constants obtained through the curve fitting of bled steam extraction pressure.*

Pressure	*m*	*N*
P_1	0.152	1.725
P_2	0.062	1.086
P_3	0.023	0.744
P_4	0.008	0.279
P_5	0.002	0.129
P_6	0.001	0.067

3.2.3 *Deaerator*

Deaerator (DR) is open type heater, the extracted steam bled is allowed to mix with feedwater and both leave at common temperature at the outlet of the heater (see Figure 7).

Similarly, for Deaerator (DR), the subscript '*i*' is fixed at 3, Energy balance

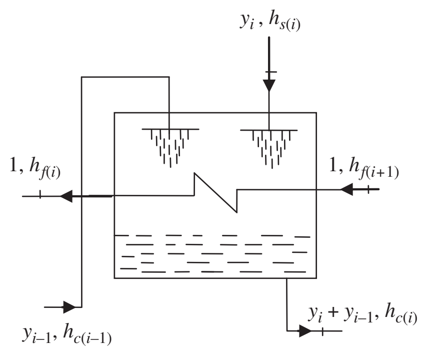

Figure 6. *Energy balance diagram for HPH-2.*

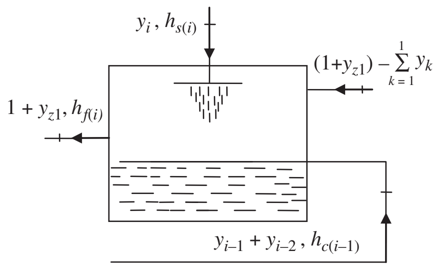

Figure 7. *Energy balance diagram for Deaerator (DR).*

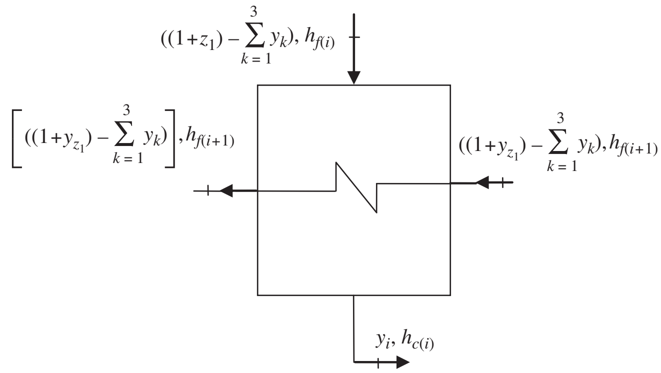

Figure 8. *Energy balance diagram for LPH-1.*

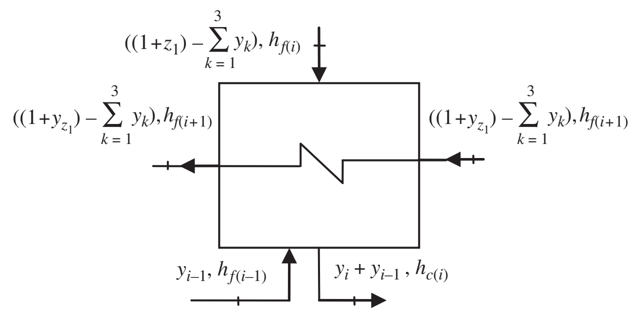

Figure 9. *Energy balance diagram for LPH-2.*

$$y_i \times h_{s(i)} + (y_{i-1} + y_{i-2}) \times h_{c(i-1)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times h_{f(i+1)} = (1 + y_{z_1}) \times h_{f(i)} \tag{21}$$

$$y_i \times h_{s(i)} + (y_{i-1} + y_{i-2}) \times h_{c(i-1)} + (1 + y_{z_1}) \times h_{f(i+1)} = (1 + y_{z_1}) \times h_{f(i)} + y_i \times h_{f(i+1)} + (y_{i-1} + y_{i-2}) \times h_{f(i+1)} \tag{22}$$

$$y_i = ((1 + y_{z_1}) \times (h_{f(i)} - h_{f(i+1)})/(h_{s(i)} - h_{f(i+1)})) - ((y_{i-1} + y_{i-2}) \times (h_{c(i-1)} - h_{f(i+1)})/(h_{s(i)} - h_{f(i+1)})) \tag{23}$$

3.2.4 *Low pressure feedwater heater (LPH-1)*

LPH-1 extracts the steam from intermediate pressure turbine. A typical intermediate pressure heater is shown in Figure 8.

Similarly, for LPH-1, the subscript i is fixed at 4,

Energy balance

$$y_i \times h_{s(i)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times h_{f(i+1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times h_{f(i)} + y_i \times h_{c(i)} \tag{24}$$

$$y_i \times (h_{s(i)} - h_{c(i)}) = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times (h_{f(i)} - h_{f(i+1)}) \tag{25}$$

$$y_i = \Bigg(\Bigg((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\Bigg) \times (h_{f(i)} - h_{f(i+1)}))/(h_{s(i)} - h_{c(i)}\Bigg) \tag{26}$$

Where

$$h_{f(i)} = h_{(i)l} - TTD \times Cpw \tag{27}$$

$$h_{c(i)} = h_{(i)l} - ETD \times Cpw \tag{28}$$

3.2.5 *Low pressure feedwater heater (LPH-2)*

The energy flow through LPH-2 is given in Figure 9.

Energy balance, for 'i' = 5:

$$y_i \times h_{s(i)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times h_{f(i+1)} + y_{i-1} \times h_{f(i-1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k\right) \times h_{f(i)} + (y_i + y_{i-1}) \times h_{c(i)} \tag{29}$$

$$y_i = \left(\left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times (h_{f(i)} - h_{f(i+1)}) \right) \Big/ (h_{s(i)} - h_{c(i)}) + (y_{i-1} \times (h_{c(i)} - h_{f(i-1)})/(h_{s(i)} - h_{c(i)})) \tag{30}$$

where

$$h_{f(i)} = h_{(i)l} - TTD \times Cpw \tag{31}$$

$$h_{c(i)} = h_{(i)l} - ETD \times Cpw \tag{32}$$

3.2.6 *Low pressure feedwater heater (LPH-3)*

The schematic energy flow diagram of LPH-3 is shown in Figure 10.

Energy balance, for 'i' = 6:

$$y_i \times h_{s(i)} + \sum_{k=4}^{5} y_k \times h_{c(i-1)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i+1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i)} + \sum_{k=4}^{6} y_k \times h_{c(i)} \tag{33}$$

$$y_i \times (h_{s(i)} - h_{c(i)}) = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times (h_{f(i)} - h_{f(i+1)}) + \sum_{k=4}^{5} y_k \times (h_{c(i)} - h_{c(i-1)}) \tag{34}$$

$$y_i = \left(\left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times (h_{f(i)} - h_{f(i+1)})/(h_{s(i)} - h_{c(i)}) \right) + \left(\sum_{k=4}^{5} y_k \times (h_{c(i)} - h_{c(i-1)})/(h_{s(i)} - h_{c(i)}) \right) \tag{35}$$

where

$$h_{f(i)} = h_{(i)l} - TTD \times Cpw \tag{36}$$

$$h_{c(i)} = h_{(i)l} - ETD \times Cpw \tag{37}$$

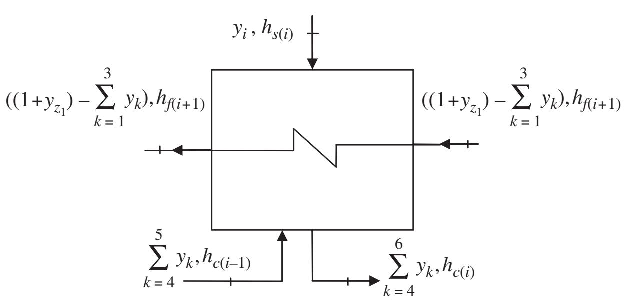

Figure 10. *Energy balance diagram for LPH-3.*

3.2.7 *Drain cooler*

Drain cooler (DC) is used in power plants if extracted steam upon condensation gets subcooled. The schematic energy flow diagram of DC is shown in Figure 11.

Energy balance, for 'i' = 7:

$$\sum_{k=4}^{6} y_k \times h_{c(i-1)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i+1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i)} + \sum_{k=4}^{6} y_k \times h_{c(i)} \tag{38}$$

$$\left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times (h_{f(i)} - h_{f(i+1)}) = \sum_{k=4}^{6} y_k \times (h_{c(i-1)} - h_{c(i)}) \tag{39}$$

$$h_{f(i)} = \left(\sum_{k=4}^{6} y_k \times (h_{c(i-1)} - h_{c(i)}) \Big/ \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \right) + h_{f(i+1)} \tag{40}$$

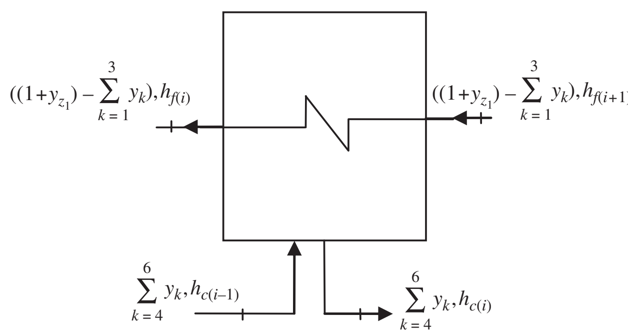

Figure 11. *Energy balance diagram for DC.*

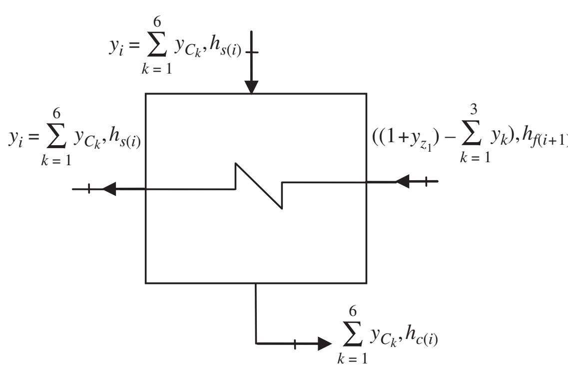

Figure 12. *Energy balance diagram for GSC.*

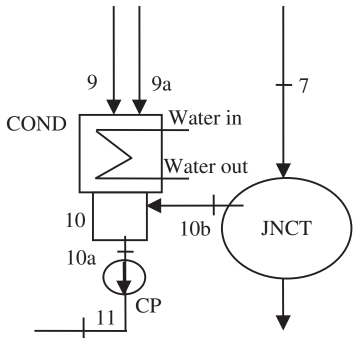

Figure 13. *Flow diagram of condenser.*

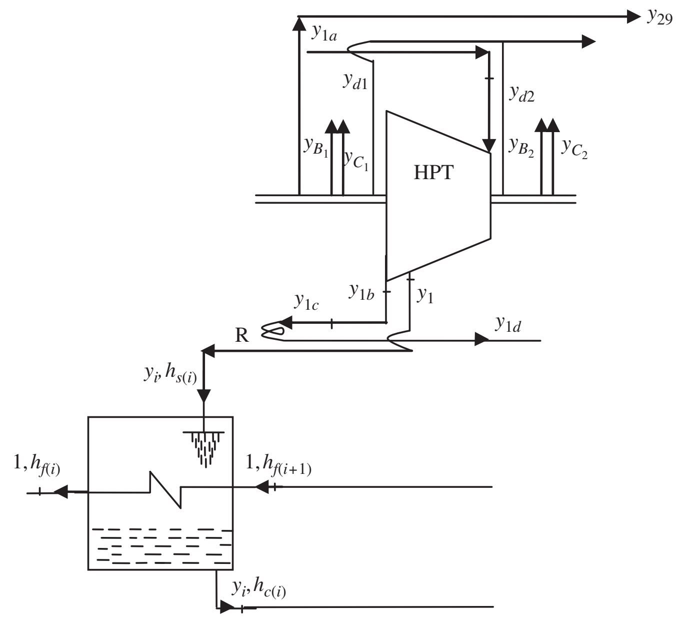

Figure 14. *High pressure turbine mass balance.*

Table 7. *Mass and energy balance of condenser.*

Mass balance	Energy balance
$y_9 + y_{9a} = y_{10}$ (44)	$y_9 h_9 + y_{9a} h_{9a} = y_{10} h_{10}$ (48)
$y_{9a} = \sum_{k=1}^{4} y_{B_k} - \sum_{k=5}^{6} y_{B_k}$ (45)	$y_9 h_9 + (\sum_{k=1}^{4} h_{B_k} y_{B_k} - \sum_{k=5}^{6} h_{B_k} y_{B_k}) = y_{10} h_{10}$ (49)
$y_{10a} = y_{10} + y_{10b}$ (46) $y_{11} = y_{10a}$ (47)	$y_{10a} h_{10a} = y_{10} h_{10} + y_{10b} h_{10b}$ (50)

3.2.8 *Gland steam condenser*

The gland steam condenser (GSC) is utilized as a low pressure noncontact feedwater heater with the discharge drainage flowing to the condenser via the condenser flash box. The gland condenser is fitted with a gland condenser extraction fan to remove any air that accumulates in the top of the gland stream condenser after the steam air mixture is separated. The energy flow through gland steam condenser is shown in Figure 12.

Energy balance, for 'i' = 8:

$$y_i \times h_{s(i)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i+1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i)} + y_i \times h_{c(i)} \tag{41}$$

$$\sum_{k=1}^{6} y_{C_k} \times h_{s(i)} + \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i+1)} = \left((1 + y_{z_1}) - \sum_{k=1}^{3} y_k \right) \times h_{f(i)} + \sum_{k=1}^{6} y_{C_k} \times h_{c(i)} \tag{42}$$

$$h_{f(i)} = \left(\sum_{k=1}^{6} y_{C_k} \times (h_{s(i)} - h_{c(i)}) / ((1 + y_{z_1}) - \sum_{k=1}^{3} y_k) \right) + h_{f(i+1)} \tag{43}$$

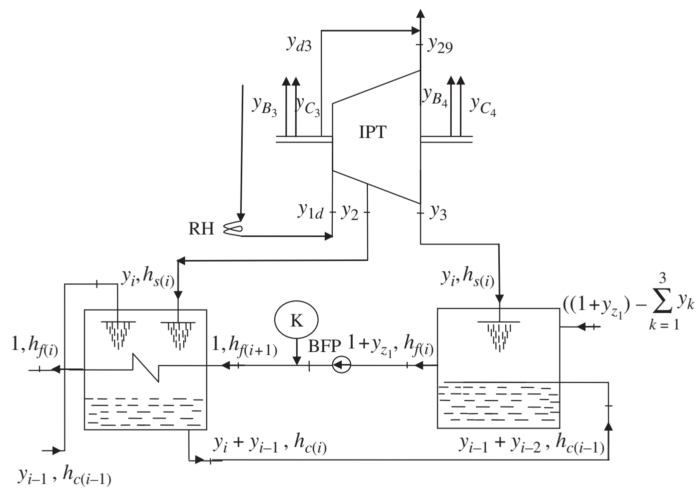

Figure 15. *Intermediate pressure turbine mass balance.*

3.3 Condenser

From the flow diagram (see Figure 13), the mass and energy balance of condenser and its junction is given as (see Table 7).

$$Q_{\text{condenser}} = x_w cp_w (T_{w,\text{out}} - T_{w,\text{in}}) = y_9 (h_9 - h_{10}) \tag{51}$$

The condition of condensed water is assumed to be saturated.

Table 8. *Energy, mass balance and work output of turbines.*

Turbine	Mass balance	Turbine work output
High pressure turbine (HPT)	$y_{1a} = \sum_{k=1}^{2} y_{B_k} + \sum_{k=1}^{2} y_{C_k} + \sum_{k=1}^{2} y_{d_k} + y_{1b}$ (52) $y_{1b} = y_{1c} + y_{1d}$ (53)	$W_{T1} = y_{1a}(h_{1a} - h_{1b})$ (58)
Intermediate pressure turbine (IPT)	$y_{1e} = \sum_{i=2}^{3} y_i + y_{32} + \sum_{k=3}^{4} y_{B_k} + \sum_{k=3}^{4} y_{C_k} + y_{d3}$ (54) $y_{32} = y_{1e} - (\sum_{i=2}^{3} y_i + \sum_{k=3}^{4} y_{B_k} + \sum_{k=3}^{4} y_{C_k}) - y_{d3}$ (55)	$W_{T2} = y_{1e}(h_{s(1e)} - h_{s(2)}) + (y_{1e} - y_2)(h_{s(2)} - h_{s(3)})$ (59)
Low pressure turbine (LPT)	$y_{29} = \sum_{i=4}^{7} y_k + \sum_{k=5}^{6} y_{B_k} - \sum_{k=5}^{6} y_{C_k} + y_9$ (56) $y_9 = y_{29} - \sum_{i=4}^{7} y_k + \sum_{k=5}^{6} y_{B_k} - \sum_{k=5}^{6} y_{C_k}$ (57)	$W_{T3} = y_{29}(h_{29} - h_{s(4)}) + (y_{29} - y_4)(h_{s(4)} - h_{s(5)}) + (y_{29} - \sum_{i=4}^{5} x_i)$ $(h_{s(5)} - h_{s(6)}) + (y_{29} - \sum_{i=4}^{6} y_i)(h_{s(6)} - h_{s(7)}) + (y_{32} - \sum_{i=4}^{7} y_i)(h_7 - h_9)$ (60)

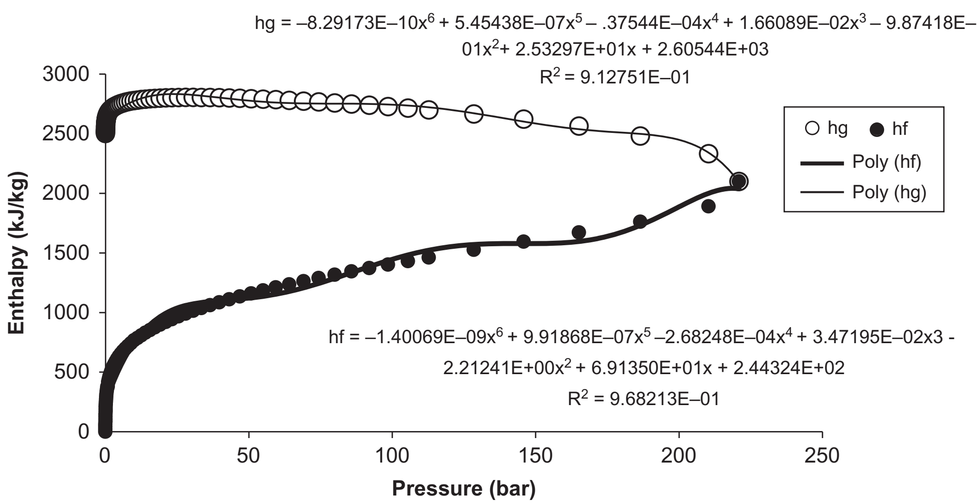

Figure 16. *Pressure-enthalpy relation of saturated water.*

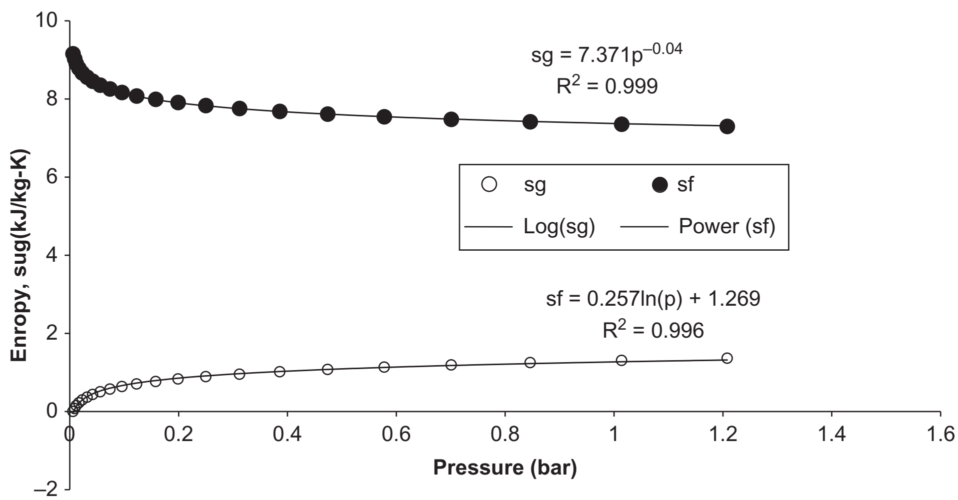

Figure 17. *Pressure-entropy diagram.*

3.4 Steam turbine

A steam turbine is one module that extracts thermal energy from pressurized steam, converts it into useful mechanical work and thus, it is one link in the chain of energy conversions with the aim of generating electrical energy. Steam turbine is condensing, tandem compounded, horizontal, reheat type, single shaft machine. It has got separate high pressure, intermediate and low-pressure parts. Mass balance of HPT and IPT are shown (see Figures 14 and 15). Energy, mass balance and work output of turbines are shown in Table 8. The notation of subscripts Bk, Ck, dk 1b, 1c and 1d represents the various states in Figures 14 and 15.

3.5 Overall thermal performance of plant

Coal consumption rate is given as

$$m_{\text{coal}} = m_{\text{uw}}(y_{1a}(h_{1a} - h_{28}) + x_{1c}(h_{1d} - h_{1c}))/(\eta_b \times \text{Coal}_{\text{GCV}}) \tag{61}$$

Plant efficiency can be evaluated in terms of plant capacity (MW) as

$$\eta_{\text{plant}} = 1000 \times \text{MW}/(m_{coal} \times \text{Coal}_{\text{GCV}}) \tag{62}$$

Here, properties modeling of working substance are given as shown in Figures 16 and 17.

The enthalpy of saturated water has been obtained from curve fit to the thermodynamic data originally by Keenan and Keyes steam tables [24] in the following form and graphical presentation is given in Figure 16.

$$\begin{aligned} \text{hu} = {} & -1.40069\ 10 \times 10^{-09}\, p^6 + 9.91868\ 10 \times 10^{-07} p^5 - 2.68248\ 10 \\ & \times 10^{-04}\, p^4 + 3.47195\ 10 \times 10^{-02}\, p^3 - 2.21241 \times p^2 \\ & + 69.1350 \times p^1 + 244.324 \end{aligned} \tag{63}$$

Enthalpy of water vapor in superheated conditions is obtained as given below

$$\begin{aligned} \text{h} = {} & \text{a}_1 + \text{a}_2\text{p}^{0.25} + \text{a}_3\text{s}^2 + \text{a}_4\text{ps}^{1.5} + \text{a}_5\text{p}^2\text{s} + \text{a}_6\text{ps}^{2.25} \\ & + \text{a}_7\text{p}^{2.05}\text{s}^{1.05} \end{aligned} \tag{64}$$

In the same manner, entropy in terms of pressure of saturated water is graphically presented in Figure 17. The equation (64) is valid in the pressure range of 0.1–175 bar and entropy in the range of 5–11.6 (kJ/kg K) with coefficient of determination of 0.9198 and with percentage error (average) of 3.0258 and RMS error of 0.1888. The values of constants are listed in Table 9. Thermodynamic properties of various points for 250 MW power cycle at design conditions are also tabulated for reference (see Table A1 in Appendix).

Table 9. *Values of constants as given in equation (64).*

Constants	Values
a_1	−1874.611439
a_2	1660.483139
a_3	48.58941216
a_4	−8.488375390
a_5	0.4763079920
a_6	1.369549685
a_7	−0.3151976165

4 RESULTS AND DISCUSSION

This proposed simulation model is based on certain assumptions for a 250 MW capacity CFPP. Empirical correlation has been modeled using design data of the plant. The validation of the simulation model can be compared with the result obtained from plant in operating conditions. Also, an attempt has been made to keep data identical for the plant thermal performance analysis. In simulation model, parameters may vary around a

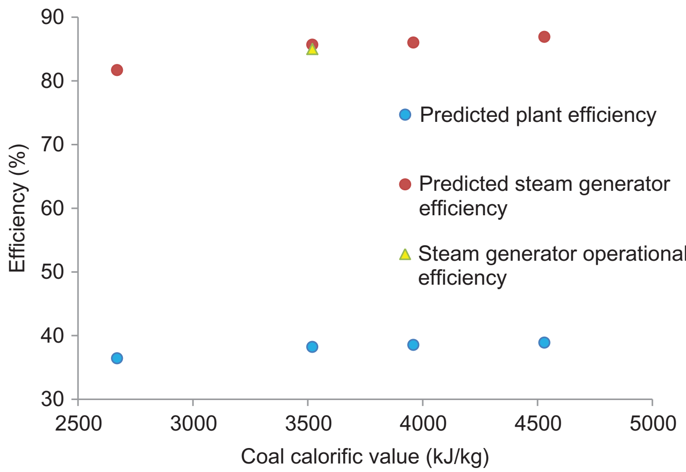

Figure 18. *Coal specimen calorific value vs. efficiency.*

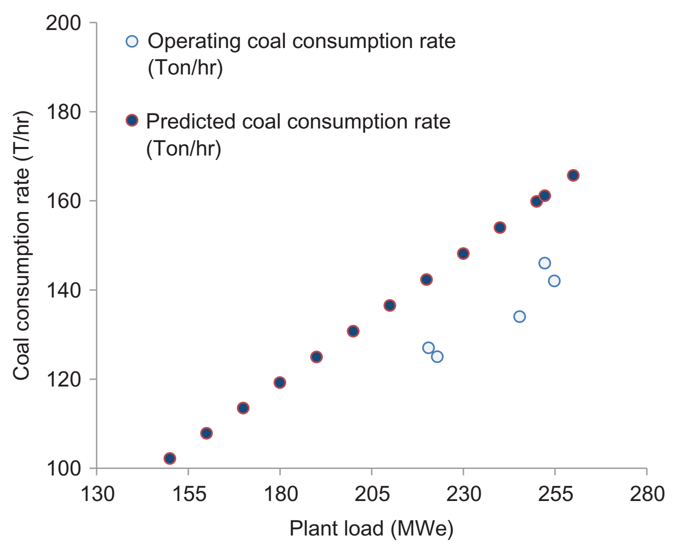

Figure 19. *Plant load vs. coal consumption rate.*

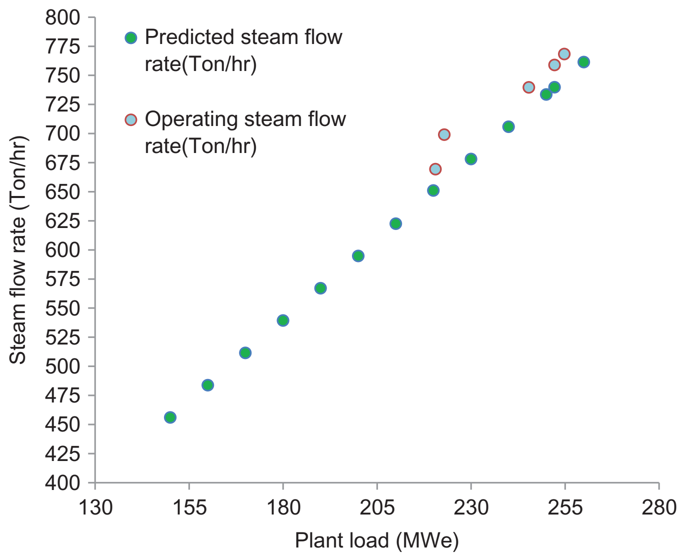

Figure 20. *Plant load vs. steam mass flow.*

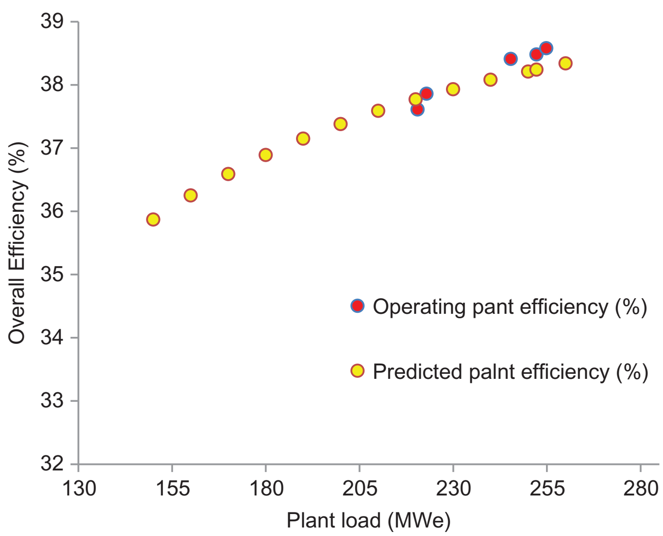

Figure 21. *Plant load vs. overall plant efficiency.*

certain baseline condition. The results obtained from simulation model and for plant in operational mode with the varying load conditions are evaluated. The overall plant efficiency, steam mass flow rate and coal consumption in plant have been achieved through the model. The comparison has been done for validation point of view as indicated through Figures 18–21. There is a crucial role of coal used in CFPP and its calorific value plays an important role in boiler combustion efficiency. Four coal specimen data have been collected from plant and the role of their calorific value has been analyzed through simulation model. In the present study, coal specimen 3 is taken as a fuel for steam generator in plant. The role of coal calorific value from various coal specimens in the evaluation of steam generator and overall plant efficiency is shown in Figure 18. Calorific value directly affects steam generator efficiency and thereby overall plant efficiency. SG and overall plant efficiency is observed to be 86.9% and 38.8%, respectively, for coal specimen-1 with coal calorific value 4530 kJ/kg. In the same manner, for coal specimen-2 with coal calorific value 3960 kJ/kg, SG and overall plant efficiency is evaluated to be 86% and 38.5%, respectively. Coal specimen-3 with calorific value 3520 kJ/kg achieves 85.6% and 38.2% SG and plant efficiency respectively. The calorific value of coal specimen-4 is much lower as compared to other three coal specimen and thereby results in lower SG and plant efficiency, which is found to be 81.6% and 36.4%, respectively obtained through simulation model. SG efficiency has been also validated by comparing it with SG efficiency in real operating condition of plant. The efficiency of SG in plant real operating condition is found to be 85% which seems to be a good agreement as achieved through simulation model for a load of 250 MW. Thus, a maximum deviation of 0.77% is observed for SG efficiency between predicted and operating data for CFPP. It is clear from Figure 18, that better coal calorific is required to achieve higher efficiency both for steam generator as well as overall plant.

A comparison of predicted and operating mass flow rate of steam through steam generator has been represented at various load conditions in plant (see Figure 20). There seems to be an increase in predicted mass flow rate of steam from 456 to 761 (ton/h) as the load varies from 150 to 260 MW, which shows a variation of 67%. It can be also seen, a maximum deviation of only 3% is observed between predicted and operating mass flow rate of steam for a load of 252 MW for a 250 MW capacity plant.

Overall efficiency analysis of any plant is the major concern for researchers. A number of factors influences plant efficiency in a direct manner. Coal calorific value is found to be one of these as discussed in this article also. Plant efficiency analysis is also successfully validated in the present work. It is clear from the Figure 21, at a load of 150 MW, plant efficiency achieved is 35.8% for selected coal specimen. As the plant load increases from 150 MW to 260 MW, plant efficiency increases from 35.8% to 38.3%, respectively which represents a maximum variation of 6.98%. In Figure 21, a comparison of predicted and operational overall plant efficiency at various load conditions is shown for selected coal specimen-3. As operating data of plant at 250 MW load was not available at the time of visit, so overall plant is compared with plant operational at 252 MW instead of 250 MW. It is clear from the Figure 21, that a maximum deviation of only 0.6% is observed between predicted and operating plant efficiency for a load of 252 MW which shows a good agreement.

5 CONCLUSIONS

A simulation model of a 250 MW capacity subcritical CFPP for the prediction of steam flow rate, coal consumption and overall plant efficiency is proposed in the present study. Thermal performance analysis of plant using a semi-empirical model has been done. The integrated model of plant with the component wise modeling for various heaters, turbine, boiler and condenser was done. Self-fitted various correlations have been also compared with Keenan and Keyes steam tables. Steam generator efficiency is evaluated using indirect method of losses. To achieve better efficiency, coal with higher calorific value is preferred. In this study, coal specimen-4 shows better steam generator efficiency as well as overall plant efficiency as compared to other specimen. During plant visit, the operational data of plant was available for coal specimen with calorific value of 3520 kJ/kg. So, coal specimen-3 has been chosen as a primary steam generator fuel. In steam generator, mass flow rate of steam improves sharply as the plant load increases. It is also concluded that, increase in plant load also increases the coal consumption for a fixed capacity CFPP for a common coal specimen used in plant. It is interesting to see the effect of load variation on overall plant efficiency. It is concluded that there is small variation in plant efficiency with the increase in load. The results obtained through the simulation model have been validated by comparing with operational plant of 250 MW capacity which shows a goods agreement.

GREEK SYMBOLS

η_{plant}	Overall plant efficiency
η_b	Steam generator efficiency
η_P	Pump efficiency

NOMENCLATURE

BP	Boiler feed pump
Ca	Unburnt carbon content in ash
CP	Condensate extraction pump
C_p	Specific heat of superheated steam in kCal/kg °C
$Coal_{GCV}$	Coal gross calorific value (kJ/kg of coal)
COND	Condenser
JNCT	Condensate collector
HPT	High pressure turbine
HPH	High pressure feedwater heater
H	Enthalpy (kJ kg^{-1})
IPT	Intermediate pressure turbine
DR	Dearator
DC	Drain cooler
$\dot{m}_f$	Coal consumption rate (ton/hr)
$\dot{m}_{uw}$	Unit mass flow rate of water (ton/hr)
P	Pressure in bar
LPH	Low pressure feedwater heater
LPT	Low pressure turbine
s	Entropy (kJ/kg K)
T_a	Ambient temperature (°C)
T_f	Flue gas temperature (°C)
$\dot{W}_P$	Power consumed by pumps (kW)
$\dot{W}_T$	Turbine work output (kW)
y_i	Fractional mass flow rate of steam at '*i*th' state
JNCT	Junction of steam and condensate collection

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APPENDIX

Table A1. *Designed thermodynamic properties of points in 250 MW power cycle (refer to Figure 1)*

Stream	m (kg/s)	t (°C)	p (bar)	h (kJ/kg)	s (kJ/kg K)
1a	204.85	537	151.98	3418.10	6.47
1b	203.40	344.2	40.13	3081.48	6.55
1c	184.09	344.2	40.13	3081.48	6.55
1d	184.09	537	36.12	3535.33	7.24
1	19.30	344.2	40.13	3081.48	6.55
2	11.66	423.4	16.78	3305.05	7.28
3	10.94	300.8	6.76	3062.22	7.31
4	7.84	188.1	2.36	2845.34	7.37
5	5.98	103.3	0.94	2684.15	7.41
6	6.12	98	0.39	2581.16	7.80
7	0.71	59.7	0.20	249.95	0.82
8	0.10	–	–	3114.97	–
9	141.38	46.7	0.10	2409.92	7.58
10a	162.94	46.3	0.10	193.84	0.65
10b	20.77	–	–	–	–
11	162.94	46.5	19.63	196.36	0.65
12	162.94	47.1	19.63	198.03	0.66
13	0.10	99.7	1.00	417.84	1.30
14	162.94	49.7	19.63	208.92	0.69
15	19.95	52.2	0.13	218.55	0.73
16	162.94	70.4	19.63	295.16	0.95
17	13.82	75.3	0.39	315.68	1.02
18	19.95	73.3	0.35	306.89	0.99
19	162.94	92.2	19.63	386.86	1.21
20	7.83	97.2	0.91	407.37	1.27
21	162.94	119.9	19.63	504.09	1.52
22	204.85	159	6.23	671.56	1.93
23	30.96	167.2	7.39	707.56	2.01
24	204.85	162.3	192.51	695.84	1.94
25	204.85	162.3	192.51	695.84	1.94
26	204.85	200.1	192.51	859.55	2.30
27	19.30	204.8	17.16	875.04	2.37
28	204.85	246	192.51	1067.63	2.72
29	204.85	300.8	6.76	3062.22	7.31
$Water_{in}$	10290.55	33	1.01	138.37	0.47
$Water_{out}$	10290.55	41.3	1.01	173.08	0.59